

Qingxiu Liu

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📍 Hong Kong, China

Education

The Chinese University of Hong Kong (CUHK)

📅 Aug 2023 – Jul 2027 (expected)

📍 Hong Kong

- Ph.D. Student in Computer Science

Institute of Computing Technology, Chinese Academy of Sciences (ICT, CAS)

📅 Sep 2020 – Jun 2023

📍 Beijing

- M.E. in Electronic and Information Engineering, GPA 3.76/4.0

Beijing University of Posts and Telecommunications (BUPT)

📅 Sep 2016 – Jun 2020

📍 Beijing

- B.E. in Information Security, GPA 3.56/4.0

Research Experiences

Memory Optimization in Mixture-of-Experts (MoE) Inference

Project Principal.

📅 Sep 2025 – Apr 2026

📍 SCITIX AI (Mentor: Dr. Yongchao He)

- Formalize expert paging, a new execution model that decouples the logical identity of experts from their physical GPU residency. Reclaim GPU memory for key-value (KV) cache by treating expert parameters as transient and streamed resources.
- Design and implement FluxMoE, an MoE inference system with three key designs: PagedTensor for virtualizing tensor addresses, a bandwidth-balanced memory hierarchy that co-optimizes loading rates across compressed GPU memory and host DRAM, and a budget-aware residency planner that dynamically selects expert residency under KV pressure.

In-Switch File-System Metadata Cache

Project Principal.

📅 Sep 2023 – Jul 2025

📍 CUHK (Supervisor: Professor Patrick P. C. Lee)

- Design and implement Fletch, an in-switch file-system metadata caching system to provide high throughput and low latencies for distributed file-system metadata services.
- Address path dependencies in in-switch caching. Design a path-aware cache admission and eviction policy for good cache efficiency, an in-switch read-write locking mechanism for efficient concurrent reads and writes, and a token-based hash collision resolution method to ensure correct metadata retrieval.

Privacy-Preserving Neural Network Inference

Project Principal.

📅 Dec 2019 – Jun 2023

📍 ICT, CAS (Supervisor: Professor Qun Huang)

- Design and implement PP-Stream, a two-party privacy-preserving neural network inference service based on homomorphic encryption and an advanced obfuscation mechanism.
- Apply stream processing architecture and two-level resource allocation to improve performance of PP-Stream. In the first level, formalize the resource allocation problem as an integer linear programming (ILP) problem. In the second level, propose an algorithm to balance the load across workers.

Two Projects on Intel SGX

Project Principal.

📅 May 2018 – Dec 2019

📍 IIIS, THU (Supervisor: Professor Wenfei Wu)

- Design and build a system called SecNF based on Intel SGX to protect user data (i.e., data to be processed within NFs) in the outsourcing service scenario.
- Design and build a trusted third-party DNS service T^2DNS based on Intel SGX to protect user privacy (i.e., association between a client IP and the queried names).

Publications

- [In Submission] FluxMoE: Decoupling Expert Residency for High-Performance MoE Serving.
Qingxiu Liu, Yongchao He, Runhan Jiang, Zion Wang, Bohan Zhao, Mi Zhang, and Patrick P. C. Lee.
- [IEEE TON] Fletch: File-System Metadata Caching in Programmable Switches. (CCF-A)
Qingxiu Liu, Jiazhen Cai, Siyuan Sheng, Yuhui Chen, Lu Tang, Zhirong Shen, and Patrick P. C. Lee.
- [IEEE ICDE 2024] PP-Stream: Toward High-Performance Privacy-Preserving Neural Network Inference via Distributed Stream Processing. (CCF-A)
Qingxiu Liu, Qun Huang, Xiang Chen, Sa Wang, Wenhao Wang, Shujie Han, and Patrick P. C. Lee.
- [IEEE ICCCN 2020] T^2DNS : A Third-Party DNS Service with Privacy Preservation and Trustworthiness. (CCF-C)
Qingxiu Liu, Wenfei Wu, Qingsong Liu, and Qun Huang.

Academic Activities

- SIGOPS ATC'26 AEC member
- Teaching Assistant at CUHK
 - CSCI4180 Introduction to Cloud Computing and Storage (Instructor: Prof. Patrick P. C. Lee): Fall 2023, Fall 2024, Fall 2025
 - AIST3020 Introduction to Computer Systems (Instructor: Prof. Patrick P. C. Lee): Spring 2024, Spring 2025

Awards

- Postgraduate Studentship in CUHK, 2023-2027
- 2nd Prize Scholarship in BUPT, 2018 and 2019
- 1st Prize in Beijing Contest District in China Undergraduate Mathematical Contest in Modeling, 2018